

Business Sales & Profit Analysis Data Analytics Project

Python | Pandas | NumPy | Matplotlib | Jupyter Notebook

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Project Overview

This project analyzes business sales data to understand overall sales performance, profitability, regional trends, and product performance.

Using Python and data visualization techniques, the project converts raw business data into clear and meaningful insights that can support better business decisions.

Tools Used: Python • Pandas • NumPy • Matplotlib



Objectives



Analyze Sales Performance

Evaluate sales trends to understand overall business performance.



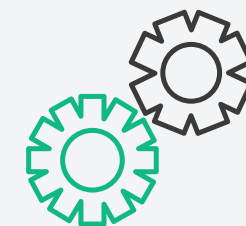
Measure Profitability

Analyze profit and profit margins to identify financial performance.



Compare Regional Performance

Identify which regions perform better in terms of sales and profitability.



Identify Key Business Drivers

Find top-performing products and unusual patterns affecting business results.



Dataset Description



The dataset contains structured information collected over a specific period. It includes multiple variables that can be analyzed to understand patterns and relationships. Each variable provides valuable information that contributes to the overall analysis.

Data Preparation



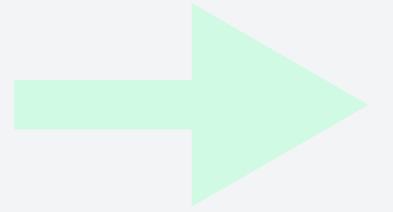
The dataset was cleaned and prepared by checking missing values, removing duplicates, correcting data types, and validating important columns. The cleaned data was then structured for accurate analysis and visualization.

```
# Check the number of rows and columns
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])

# Display column names
print("\nColumn Names:")
print(df.columns.tolist())

# Display data types and non-null counts
print("\nDataset Information:")
df.info()
```

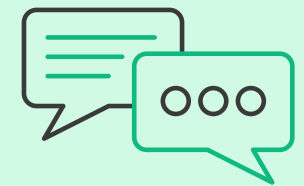
Exploration Process



```
# Statistical summary of numerical columns  
print("Statistical Summary:")  
display(df.describe())
```



The cleaned dataset was explored using descriptive statistics, grouping, aggregation, and visualizations to identify sales trends, profit patterns, regional performance, top products, and unusual values.



Analytical Approach



Descriptive Analysis

Summarize sales and profit to understand overall business performance.



Regional Analysis

Compare sales, profit, and margins across different regions.



Product Analysis

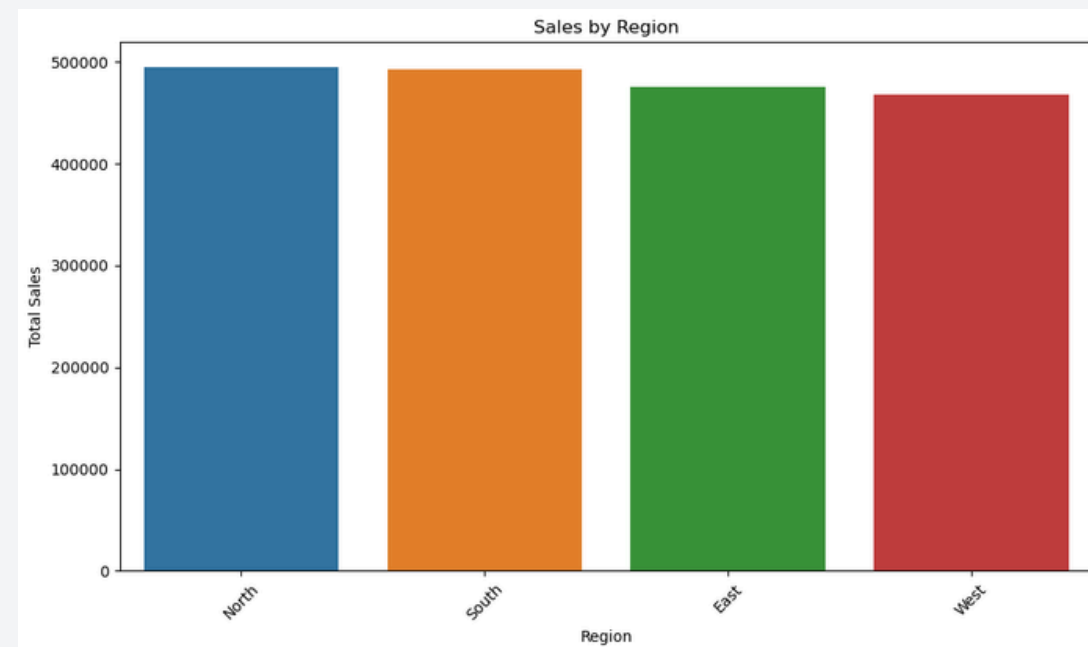
Identify top-performing products based on sales.



Outlier Analysis

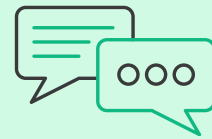
Detect unusual values that may affect business performance.

Visualization Strategy

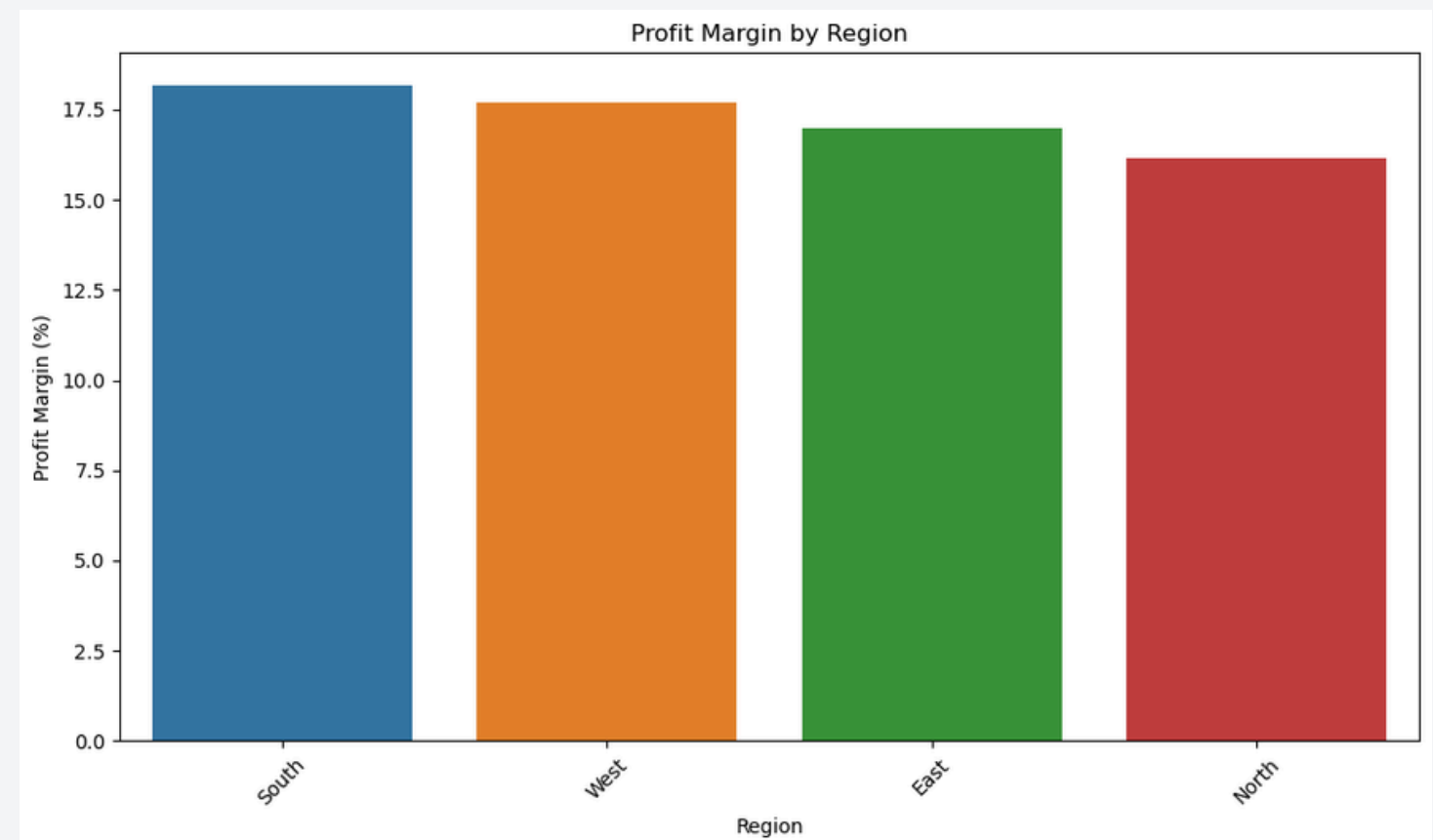
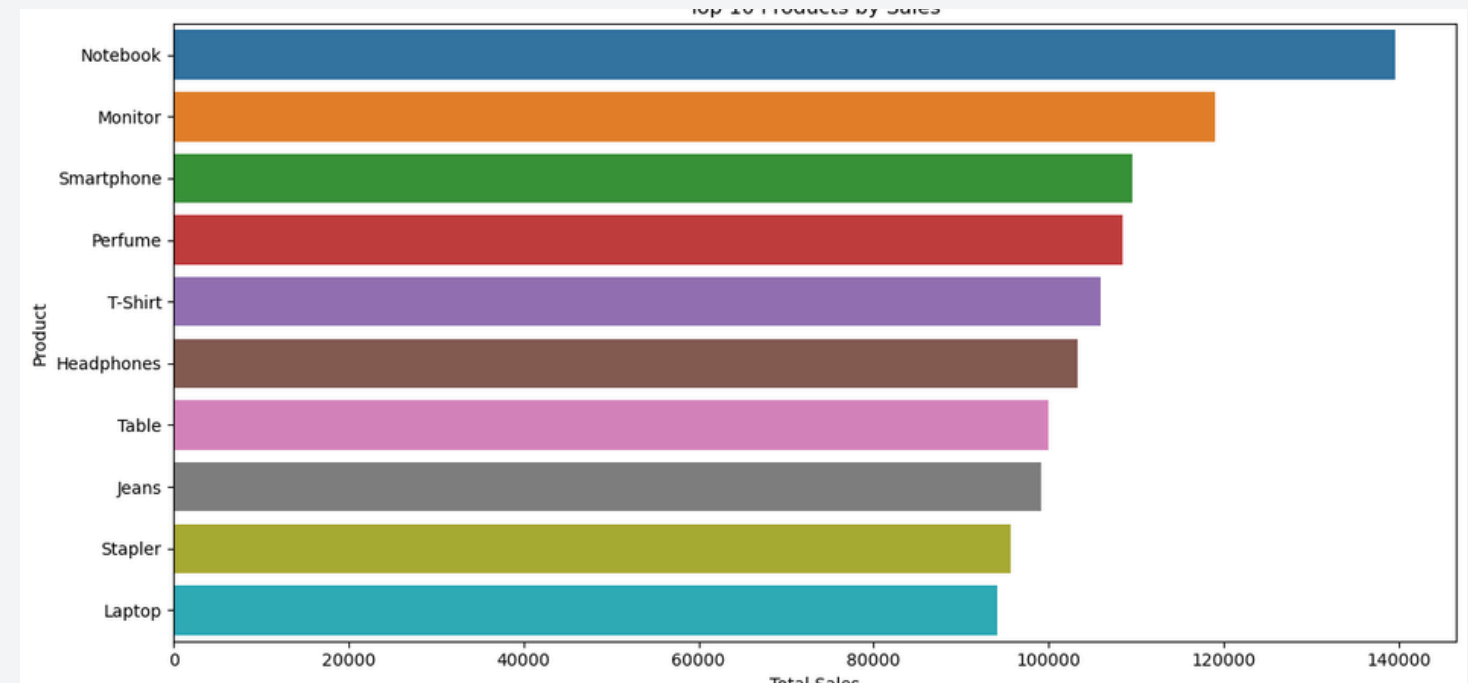


Charts and graphs were used to present sales, profit, regional performance, and product trends clearly, making complex data easier to understand and compare.

Observed Trends



The analysis revealed clear differences in sales, profitability, regional performance, and product demand, highlighting the key factors driving overall business performance.



Interpretation of Results



The results show that sales volume alone does not determine business success. Profit margins, regional performance, and product demand provide deeper insights into business efficiency. These findings help identify high-performing areas, improvement opportunities, and potential risks for better decision-making.



Practical Value

The analysis helps businesses identify profitable regions, focus on high-performing products, optimize costs, and make data-driven decisions to improve overall performance.





Summary

This project transformed raw business data into meaningful insights by analyzing sales, profit, regional performance, product trends, and outliers. The analysis supports better understanding of business performance and data-driven decision-making.



“Turning Data Into Decisions.”

Thank You
Presented by:
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Thank You

Questions are welcome



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